

MAT 528 HW05

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Problem 1 (Munkres 24.1). (a) Show that no two of the spaces $(0, 1)$, $(0, 1]$, and $[0, 1]$ are homeomorphic.

(b) Suppose that there exist embeddings $f: X \rightarrow Y$ and $g: Y \rightarrow X$. Show by means of an example that X and Y need not be homeomorphic.

(c) Show $\mathbb{R}^n$ and $\mathbb{R}$ are not homeomorphic if $n > 1$.

Solution. (a) First, let A be either of $(0, 1]$ or $[0, 1]$ and suppose that $h: A \rightarrow (0, 1)$ is a homeomorphism. Then, $A \setminus \{1\}$ is connected (because it is either $(0, 1)$ or $[0, 1)$), so

$$h(A \setminus \{1\}) = h(A) \setminus \{h(1)\} = (0, 1) \setminus \{h(1)\}$$

is also connected (where the first equality is because h is bijective). However, $(0, 1) \setminus \{h(1)\} = (0, h(1)) \cup (h(1), 1)$ is disconnected because it is the union of nonempty disjoint open sets $(0, h(1))$ and $(h(1), 1)$. Therefore, $(0, 1)$ is not homeomorphic to either of $(0, 1]$ or $[0, 1]$.

Now, let $g: [0, 1] \rightarrow (0, 1]$ be a homeomorphism. Then

$$g([0, 1)) = (0, 1] \setminus \{g(1)\}$$

is connected, which means $g(1) = 1$, since otherwise this set is the union of the nonempty disjoint open sets $(0, g(1)) \cup (g(1), 1]$ (they are open in $(0, 1]$). Similarly

$$g((0, 1]) = (0, 1] \setminus \{g(0)\}$$

is connected, also implying $g(0) = 1$; this contradicts the fact that g must be bijective to be a homeomorphism, so $(0, 1]$ and $[0, 1]$ are not homeomorphic.

(b) $X = (0, 1)$ and $Y = [0, 1]$ are not homeomorphic, but X can be embedded in Y by the inclusion map $f(x) = x$, and Y can be embedded in X by the map $g: Y \rightarrow X$ defined as

$$g(t) = \frac{1 + 2t}{4};$$

this map is continuous and injective, and its inverse $g^{-1}: g(X) \rightarrow X$ is continuous because it is $g^{-1}(t) = \frac{4t-1}{2}$.

(c) Let $h: \mathbb{R}^n \rightarrow \mathbb{R}$ be a homeomorphism. Then, $\mathbb{R}^n \setminus \{0\}$ is connected because it is path connected: for most $x, y \in \mathbb{R}^n$, the segment between x and y is a path in $\mathbb{R}^n \setminus \{0\}$ connecting the two points; the only exception is if the segment contains 0, in which case we can take the path consisting of the two segments between x and p and between p and y , where p is a point not on the line through x and y . This means $h(\mathbb{R}^n \setminus \{0\}) = \mathbb{R} \setminus \{h(0)\}$ is connected, but this is false, because $\mathbb{R} \setminus \{h(0)\}$ is the union of the nonempty disjoint open sets $(-\infty, h(0)) \cup (h(0), \infty)$, so h cannot exist. □

Problem 2 (Munkres 24.2). Let $f: S^1 \rightarrow \mathbb{R}$ be a continuous map. Show that there exists a point $x \in S^1$ such that $f(x) = f(-x)$.

Solution. Define $g: S^1 \rightarrow \mathbb{R}$ by $g(x) = f(x) - f(-x)$; we want to show that there exists $x_0 \in S^1$ such that $g(x_0) = 0$. We can see that $g(-x) = f(-x) - f(x) = -g(x)$. If $g(x) = 0$ we are done; otherwise, 0 lies between $g(x)$ and $g(-x)$ so, since S^1 is connected, the intermediate value theorem tells us that there exists $x_0 \in S^1$ such that $g(x_0) = 0$, as desired. $\square$

Problem 3 (Munkres 24.3). Let $f: X \rightarrow X$ be continuous. Show that if $X = [0, 1]$, there is a point x such that $f(x) = x$. The point x is called a **fixed point** of f . What happens if X equals $[0, 1)$ or $(0, 1)$?

Solution. If $X = [0, 1]$, let $g(x) = f(x) - x$. Then $g(0) = f(0) - 0 \geq 0 - 0 = 0$, while $g(1) = f(1) - 1 \leq 1 - 1 = 0$, so 0 is between $g(0)$ and $g(1)$; by the intermediate value theorem, there exists $x_0 \in X$ such that $g(x_0) = 0$, which implies $f(x_0) = x_0$.

If $X = (0, 1)$ or $[0, 1)$, then $f(x) = \frac{1}{2}(x + 1)$ satisfies

$$f(x) = \frac{1}{2}x + \frac{1}{2} \in (x, 1)$$

for all $x < 1$, so this f doesn't have a fixed point and the statement fails to hold. $\square$

Problem 4 (Munkres 24.11). If A is a connected subspace of X , does it follow that $\text{Int } A$ and ∂A are connected? Does the converse hold?

Solution. Both the statement and its converse are false.

First, $X = [0, 1]$ is connected but has boundary $\{0, 1\}$, which is disconnected (because it inherits the discrete topology as a subspace of $\mathbb{R}$), so the boundary of a connected set may be disconnected. To show that the interior of a connected set may be disconnected, let $X \subseteq \mathbb{R}^2$ be the union of the two closed disks of radius 1 centered at $(-1, 0)$ and $(1, 0)$ so the disks intersect only at the point $(0, 0)$; this nonempty intersection implies that X is connected because the disks themselves are connected. However, the interior of X doesn't include $(0, 0)$ (and of course no other point on the y -axis since X only intersects the y -axis at $(0, 0)$) because any open ball around $(0, 0)$ contains some point $(0, r)$ for $r \neq 0$, so X has the separation $(X \cap H_-) \sqcup (X \cap H_+)$, where $H_- \subseteq \mathbb{R}^2$ is the open set consisting of the points with a negative x -coordinate and $H_+ \subseteq \mathbb{R}^2$ is the open set consisting of the points with a positive x -coordinate.

For the converse, we note that $\mathbb{Q}$ has empty interior (since any nonempty open interval contains an irrational number — the irrational numbers are dense in $\mathbb{R}$) and its boundary is $\overline{\mathbb{Q}} = \mathbb{R}$ (minus the interior of $\mathbb{Q}$, but that's empty), so the interior and boundary of $\mathbb{Q}$ are connected. However, $\mathbb{Q}$ itself is disconnected, for example having the separation $(\mathbb{Q} \cap (-\infty, \sqrt{2})) \sqcup (\mathbb{Q} \cap (\sqrt{2}, \infty))$. $\square$

Problem 5 (Munkres 25.4). Let X be locally path connected. Show that every connected open set in X is path connected.

Solution. By Munkres's Theorem 25.3, since X is locally path connected, each path component of an open subset $U \subseteq X$ is open in X . If U has multiple path components, then let P be one of the path components and Q be the union of the other path components; both P and Q are open in U , nonempty, disjoint, and have union U (because the path components of U partition U), so U is disconnected. Contrapositively, if U is connected, then it must have only one path component, i.e. U is also path connected. $\square$

Problem 6 (Munkres 25.5). Let X denote the rational points of the interval $[0, 1] \times \{0\}$ of $\mathbb{R}^2$. Let T denote the union of all line segments joining the point $p = (0, 1)$ to points of X .

- (a) Show that T is path connected, but is locally connected only at the point p .
- (b) Find a subset of $\mathbb{R}^2$ that is path connected but is locally connected at none of its points.

Solution. (a) T is path connected because given x and y in T , there is a path traversing $x \rightarrow p \rightarrow y$ via the segments joining x to p and p to y . By the same argument, the intersection $T \cap B_\varepsilon(p)$ with an open ball centered at p is path connected, since if $x \in T \cap B_\varepsilon(p)$ then the segment from p to x is contained in $T \cap B_\varepsilon(p)$ (if this weren't true, then, noting that the segment is in T , we would have a contradiction to the fact that $B_\varepsilon(p)$ is convex). Thus, for any open set $U \subseteq T$ containing p , taking $\varepsilon > 0$ small enough so that $T \cap B_\varepsilon(p) \subseteq U$ shows that T is locally path connected at p .

Now consider $T^* = T \setminus \{p\}$, which is an open subset of T . Let $f: T^* \rightarrow \mathbb{Q} \cap [0, 1]$ be the map such that points on the segment in T^* connecting $(0, 1)$ to a rational point $(r, 0)$ (but of course not containing $(0, 1)$ because we excluded it from T^*) for $r \in \mathbb{Q} \cap [0, 1]$ are sent to r . This map is continuous because the pre-image of the open ball $(q_1, q_2) \cap \mathbb{Q}$ is the union of the segments corresponding to the elements of that set, which make up the open subset

$$T^* \cap \left\{ (x, y) : 1 - \frac{x}{q_1} < y < 1 - \frac{x}{q_2} \right\}$$

(intersecting T^* with the region between the two lines).

For each point x in T^* , a neighborhood U of that point intersects infinitely many of the segments that comprise T^* , so $f(U)$ is an infinite subset of $\mathbb{Q} \cap [0, 1]$. With the subspace topology, $\mathbb{Q} \cap [0, 1]$ has only the singleton sets as its connected components, so $f(U)$ is disconnected, hence so is U . Since no open subset of x is connected, T^* , hence T , is not locally connected at x .

(b) Rotate T by π about $(\frac{1}{2}, \frac{1}{2})$ to get T' , with $p' = (1, 0)$ being the image of p under the rotation. The map f and the corresponding map $f': T'^* \setminus \{p'\} \rightarrow \mathbb{Q} \cap [0, 1]$ given by applying f to the pre-image of each point in $(T')^*$ under the rotation are both continuous, and f, f' agree on $T^* \cap (T')^*$ because they are both 1 on this segment, so by the pasting lemma $g: T^* \cup (T')^* = T \cup T' \rightarrow \mathbb{Q} \cap [0, 1]$ defined by $g|_{T^*} = f$ and $g|_{(T')^*} = f'$ is continuous. Then, one can apply the argument of (a) to get that any neighborhood of any point of f is disconnected, despite path connectedness holding since the path connected sets T and T' intersect on the segment between p and p' which is path connected. $\square$